10.16 The $i\epsilon$ Prescription and Logarithms in Loop Integrals

Introduction: The Problem of the Imaginary Part

When calculating loop integrals in Quantum Field Theory, one frequently encounters logarithmic terms that depend on Mandelstam variables, such as $\log(-s/m^2)$ or $\log(t/m^2)$. A critical question arises when these kinematic invariants are in their physical regions (e.g., $s > 0$ and $t < 0$), as the argument of the logarithm becomes negative. This introduces an imaginary part of the form $\pm i\pi$. These logarithms come from the Feynman-parameter denominator $D(x_i; k_j)$, which always contains a $+m^2$ term (in our metric), so the $-i\epsilon$ can be restored next to it.[2]

The sign is not arbitrary; it is unambiguously fixed by the fundamental principles of causality. This physical requirement is mathematically encoded in the Feynman $i\epsilon$ prescription (§10.8). This section explains how this one fundamental rule determines the correct imaginary parts for all resulting functions.

The Fundamental Rule: Causality and the Feynman Propagator

The origin of all $i\epsilon$ rules in a scattering amplitude can be traced back to a single source: the Feynman propagator. For a scalar particle of mass m, the propagator line in a Feynman diagram carries the factor

$$\frac{1}{i}\,\tilde{\Delta}(p) = \frac{-i}{p^2 + m^2 - i\epsilon} \quad \text{where } \epsilon \to 0^+, \tag{10.16.1}$$

the mostly-plus form of the familiar $i/(p^2 - m^2 + i\epsilon)$. This $-i\epsilon$ (which can be thought of as giving the mass a small negative imaginary part, $m^2 \to m^2 - i\epsilon$) is the precise mathematical statement of causality. It ensures that positive-energy states propagate forward in time.

Every other prescription for handling singularities in the final, integrated amplitude is a **consequence** of applying this fundamental rule to all propagators in the loop *before* integration.

[2] The m^2 in the denominator is nothing but a dimensionful quantity to make the argument dimensionless, so it doesn't come with a $-i\epsilon$.

The Consequence: A "Simple Rule" for Kinematic Invariants

When we perform the loop integration (e.g., using Feynman parameters), the $i\epsilon$ terms from all internal propagators combine, and together they make the integrated amplitude $I(s,t,u,\dots)$ an analytic function of its kinematic variables.
This leads to a simple, practical shortcut for evaluating the final expressions, which we can call the "Simple Rule":

> To find the correct imaginary part of a final amplitude, replace all independent kinematic invariants x (like s, t, and u) with $\mathbf{x + i\epsilon}$.

This rule is the *result* of the fundamental $m^2 \to m^2 - i\epsilon$ prescription, not a new or independent assumption. After Feynman parametrization the denominator has the form $D = \sum_i x_i m_i^2 - \sum_{i<j} x_i x_j\, s_{ij} - i\epsilon$ (for example $D = m^2 - x(1-x)s - i\epsilon$ for a bubble), with Feynman parameters $x_i \geq 0$, $\sum_i x_i = 1$, and s_{ij} the invariants (defined to be positive in their own physical channel). Each invariant appears multiplied by a non-positive product of Feynman parameters, so the $-i\epsilon$ in D can be absorbed as $s_{ij} \to s_{ij} + i\epsilon$.

Applying the Simple Rule to Logarithms

We use the standard "principal branch" convention for the complex logarithm, where the argument $\mathrm{Arg}(z)$ is defined to be in the range $(-\pi, \pi]$. The imaginary part of the log is determined by the quadrant of its argument.

Case 1: $\log(t/m^2)$

In this region, t is the (spacelike) momentum transfer, so $t < 0$. Let $t = -|t|$.

- **Apply Rule:** We replace $t \to t + i\epsilon = -|t| + i\epsilon$.
- **Evaluate Log:** $\log\left(\frac{t+i\epsilon}{m^2}\right) = \log\left(\frac{-|t|+i\epsilon}{m^2}\right)$.
- **Analysis:** The argument $z = (-|t| + i\epsilon)/m^2$ has a negative real part and a small positive imaginary part. It is in the **second quadrant**.
- **Result:** $\mathrm{Arg}(z) = +\pi$. Therefore:

$$\log(t/m^2) \to \log\left(\frac{|t|}{m^2}\right) + i\pi$$

Case 2: $\log(-s/m^2)$

In this region, s is the (timelike) center-of-mass energy squared, so $s > 0$.

- **Apply Rule:** We replace $s \to s + i\epsilon$.
- **Evaluate Log:** $\log\left(\frac{-(s+i\epsilon)}{m^2}\right) = \log\left(\frac{-s-i\epsilon}{m^2}\right)$.
- **Analysis:** The argument $z = (-s - i\epsilon)/m^2$ has a negative real part and a small negative imaginary part. It is in the **third quadrant**.
- **Result:** $\text{Arg}(z) = -\pi$. Therefore:

$$\log(-s/m^2) \to \log\left(\frac{s}{m^2}\right) - i\pi$$

This imaginary part is physically crucial. It is related via the optical theorem to the discontinuity of the amplitude across its branch cut, which corresponds to the probability of the intermediate particles going on-shell.

